

NAGANO, Japan

WMO: 476100

Lat: 36.6626N

Lon: 138.1920E

Elev: 420

StdP: 96.38

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-5.5	-10.5	1.6	-2.0	-9.2	1.8	-2.0	8.2	2.8	7.1	2.3	1.4	270	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	33.2	23.8	31.8	23.4	30.3	22.8	24.7	30.8	24.1	29.7	23.5	28.6	3.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	18.7	27.2	22.5	18.1	26.8	22.0	17.5	26.5	77.6	30.9	74.9	29.9	72.5	28.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.9	6.1	-9.2	35.6	1.1	1.0	-10.0	36.3	-10.6	36.9	-11.3	37.4	-12.1	38.1		
(5)			-9.7	25.8	1.0	0.7	-10.4	26.3	-11.0	26.7	-11.6	27.1	-12.3	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-0.3	0.7	4.7	11.1	17.0	20.9	25.1	26.2	21.6	15.1	8.3	2.6		
	DBStd	9.66	2.40	2.90	3.52	4.01	3.07	2.53	2.49	2.44	3.42	3.32	3.20	2.95		
	HDD10.0	1087	321	261	168	34	1	0	0	0	0	2	72	229		
	HDD18.3	2690	579	494	422	218	61	7	0	0	9	111	302	487		
	CDD10.0	2116	0	0	5	67	218	328	468	503	348	159	19	1		
	CDD18.3	677	0	0	0	1	20	84	209	244	107	10	0	0		
	CDH23.3	4820	0	0	0	19	175	429	1538	2022	603	33	0	0		
	CDH26.7	1604	0	0	0	2	28	83	544	779	164	4	0	0		
(14) Wind	WSAvg	2.5	2.0	2.3	2.8	3.0	3.0	2.6	2.3	2.5	2.6	2.4	2.2	2.0		
(15) Precipitation	PrecAvg	1163	83	68	76	69	80	117	136	117	146	110	78	83		
	PrecMax	1395	160	105	119	148	199	180	342	203	258	339	146	155		
	PrecMin	753	41	38	30	21	38	71	34	24	30	27	31	47		
	PrecStd	136	27	17	23	30	37	32	69	46	54	65	25	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	13.4	19.6	26.0	29.5	30.8	34.4	35.0	32.5	26.7	19.5	14.1		
		MCWB	5.1	7.8	10.0	14.1	17.5	20.5	23.8	24.0	22.8	19.4	13.1	9.8		
	2%	DB	6.7	10.0	16.4	22.7	27.1	28.8	32.9	33.6	30.5	23.7	17.1	11.0		
		MCWB	3.3	5.4	8.6	12.7	16.4	19.8	23.8	24.0	22.2	17.5	11.9	7.3		
	5%	DB	5.0	7.5	13.5	20.5	25.3	27.4	31.4	32.3	28.5	22.0	15.4	9.2		
		MCWB	2.2	3.7	7.4	11.7	15.4	19.5	23.4	23.8	21.5	16.5	11.3	6.2		
	10%	DB	3.6	5.6	11.1	18.3	23.4	25.9	29.8	30.9	26.7	20.2	13.8	7.5		
		MCWB	1.3	2.6	6.4	11.1	14.9	19.0	22.9	23.3	20.8	15.4	10.4	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	9.2	11.8	16.7	19.7	23.1	25.5	25.7	23.9	20.3	14.7	10.7		
		MCDB	8.7	12.4	16.9	21.0	25.4	27.6	32.2	32.6	29.9	25.0	16.9	12.8		
	2%	WB	3.9	5.8	9.7	14.4	18.1	21.8	24.6	25.0	23.0	18.7	13.2	8.2		
		MCDB	6.2	8.9	14.0	20.3	24.3	25.9	30.7	31.2	28.2	22.5	15.8	10.5		
	5%	WB	2.5	4.1	8.1	13.1	16.9	20.9	24.0	24.4	22.2	17.4	12.0	6.4		
		MCDB	4.5	6.9	12.8	18.5	22.4	25.1	29.5	30.1	27.0	20.8	14.5	8.9		
	10%	WB	1.5	2.8	6.6	11.8	15.9	20.0	23.3	23.8	21.5	16.2	10.8	4.9		
		MCDB	3.3	5.4	10.8	16.9	21.6	24.3	28.4	29.1	25.8	19.4	13.5	7.2		
(35) Mean Daily Temperature Range	MDBR		6.9	8.1	10.0	11.5	11.4	9.2	8.5	8.8	8.2	8.7	9.1	7.5		
	5% DB	MCDBR	9.2	11.4	15.0	15.1	14.7	11.6	10.6	10.5	9.9	10.3	11.0	9.9		
		MCWBR	6.2	7.3	8.3	7.2	6.2	4.7	3.9	3.6	3.8	4.9	6.4	6.6		
	5% WB	MCDBR	8.5	10.7	13.3	12.3	11.7	9.0	9.7	9.7	8.8	8.3	9.2	9.3		
		MCWBR	6.1	7.3	8.1	6.6	5.7	4.0	3.8	3.6	3.6	4.4	5.9	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.381	0.421	0.487	0.528	0.547	0.581	0.569	0.569	0.496	0.442	0.398	0.380		
	taud		1.989	1.902	1.835	1.839	1.850	1.836	1.933	1.928	2.078	2.136	2.178	2.094		
	Ebn at Noon		767	785	771	766	759	732	737	725	759	762	754	743		
	Edh at Noon		134	165	194	204	205	208	188	185	151	130	111	114		
(44) All-Sky Solar Radiation	RadAvg		2.01	2.81	3.77	4.69	5.29	4.89	4.71	4.77	3.81	3.11	2.28	1.79		
	RadStd		0.14	0.20	0.26	0.38	0.54	0.33	0.69	0.48	0.31	0.29	0.18	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +0.50	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page